DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTETC602 - DIGITAL COMMUNICATION

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 In digital transmission, the modulation technique that requires the minimum bandwidth is:
A. PCM
B. PAM
C. DPCM
D. Delta modulation

2 What is the key drawback of Delta Modulation (DM)?
A. Complexity
B. High bandwidth requirement
C. Slope overload distortion
D. High cost

3 Which type of multiplexing technique is most commonly used in digital transmission systems?
A) Frequency Division Multiplexing (FDM)
B) Time Division Multiplexing (TDM)
C) Code Division Multiplexing (CDM)
D) Wavelength Division Multiplexing (WDM)

4 Intersymbol interference (ISI) primarily occurs due to:
A) Noise
B) Bandwidth limitations of the channel
C) Frame synchronization errors
D) Multiplexer faults

5 **White noise has a power spectral density that is:**
A) Constant over all frequencies
B) Zero at DC
C) High-pass in nature
D) Band-limited

6 **A random process is:**

A) A deterministic function of time
B) A single function with no variability
C) A collection of time functions indexed by a random variable
D) A fixed number sequence

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_____is a multilevel modulation in which four phase shift are used for representing four different symbols.

A. QPSK
B. BFSK
C. BPSK
D. 8-PSK

8 In BPSK, the ________ of constant amplitude carrier signal is switched between two values according to the two possible values.

A. Amplitude
B. Phase
C. Frequency
D. Angle

9 Jamming refers to:

A) Interference caused by multiple users sharing the channel
B) Deliberate interference to degrade signal reception
C) Reflection of signals from buildings
D) Compression of signal bandwidth

10 In Direct Sequence Spread Spectrum (DSSS), the data signal is:

A) Multiplied by a high-rate PN sequence
B) Frequency modulated
C) Time-division multiplexed
D) Encoded with parity bits

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In the context of digital communications, what is the primary benefit of using a matched filter?

A. It reduces inter-symbol interference.
B. It requires less power for the same performance.
C. It increases the transmission bandwidth.
D. It enhances the phase stability of the received signal.

12 In the block diagram of a digital communication system, the source encoder is responsible for:

A) Converting analog signal to binary
B) Reducing redundancy in source data
C) Modulating the signal
D) Error correction

Q2. Solve the following.

A) With a neat block diagram, explain the PCM generation process. 6

B) A Television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that the number of quantization levels is 512. Determine i). Codeword length? ii). Transmission Bandwidth? iii). Final Bit rate? iv). Output SNR ratio? 6

Q3. Solve the following.

A) Define equalization. Explain with the help of a diagram how equalizers help reduce the effect of intersymbol interference (ISI) in a digital communication system. 6

B) Explain the concept of digital multiplexing. With a neat diagram 6

Q4. Solve Any Two of the following.

A) Define a random process. Explain its mathematical representation and components with suitable examples. 6

B) Define delta modulation. With the help of a block diagram, explain the working of a delta modulation transmitter and receiver. 6

C) Explain the transmission of a random process through an LTI system. Provide the mathematical expression for the output process and discuss how statistical properties of the input affect the output. 6

Q5. Solve Any Two of the following.

A) Derive the properties of matched filter. 6

B) What are Pseudo Noise (PN) sequences? Explain the generation of PN sequences using Linear Feedback Shift Registers (LFSRs) with an example. 6

C) Describe the generation and detection process of BPSK. Provide a block diagram and explain each block in detail. 6

Q6. Solve Any Two of the following.

A) Explain the QPSK generation process with the help of a block diagram and draw the corresponding waveforms. 6

B) 'What is Frequency Hop Spread Spectrum? Explain its working with the help of a block diagram 6

C) Explain the Concept of jamming in detail 6

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